

Multiple Choice (20 marks)

1. A 2.35-gram sample was dissolved in water, and the chloride ions were precipitated by adding silver nitrate ($\text{Ag}^+ + \text{Cl}^- \rightarrow \text{AgCl}$). If 0.435 g of precipitate was...
 - (a) 10.80%
 - (b) 4.60%
 - (c) 43.50%
 - (d) 18%
2. A 2.0 L flask holds 0.40 g of helium gas. If the helium is evacuated into a larger container while the temperature is held constant, what will the effect on...
 - (a) It will remain constant as the number of helium molecules does not change.
 - (b) It will decrease as the gas will be more ordered in the larger flask.
 - (c) It will decrease because the molecules will collide with the sides of the larger flask less often than they did in the smaller flask.
 - (d) It will increase as the gas molecules will be more dispersed in the larger flask.
3. The electrolysis of molten magnesium bromide is expected to produce
 - (a) magnesium at the anode and bromine at the cathode
 - (b) magnesium at the cathode and bromine at the anode
 - (c) magnesium at the cathode and oxygen at the anode
 - (d) bromine at the anode and hydrogen at the cathode
4. What is the general relationship between temperature and entropy for diatomic gases?
 - (a) They are completely independent of each other; temperature has no effect on entropy.
 - (b) There is a direct relationship, because at higher temperatures there is an increase in energy dispersal.
 - (c) There is an inverse relationship, because at higher temperatures substances are more likely to be in a gaseous state.
 - (d) It depends on the specific gas and the strength of the intermolecular forces between individual molecules.
5. $\text{NH}_4^+(\text{aq}) + \text{NO}_2^-(\text{aq}) \rightarrow \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$ Increasing the temperature of the above reaction will increase the rate of reaction. Which of the following is NOT a...
 - (a) The reactants will be more likely to overcome the activation energy.

- (b) The number of collisions between reactant molecules will increase.
 - (c) A greater distribution of reactant molecules will have high velocities.
 - (d) Alternate reaction pathways become available at higher temperatures.
6. Three half-lives after an isotope is prepared,
- (a) 25% of the isotope is left
 - (b) 25% of the isotope decayed
 - (c) 12.5% of the isotope is left
 - (d) 12.5% of the isotope decayed
7. Which of the following processes is an irreversible reaction?
- (a) $\text{CH}_4(\text{g}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
 - (b) $\text{HCN}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{CN}^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$
 - (c) $\text{Al}(\text{NO}_3)_3(\text{s}) \rightarrow \text{Al}^{3+}(\text{aq}) + 3\text{NO}_3^-(\text{aq})$
 - (d) $2\text{Ag}^+(\text{aq}) + \text{Ti}(\text{s}) \rightarrow 2\text{Ag}(\text{s}) + \text{Ti}^{2+}(\text{aq})$
8. When potassium perchlorate, KClO_4 , dissolves in water, the temperature of the resultant solution is lower than the initial temperature of the components....
- (a) This is a spontaneous process because it is exothermic.
 - (b) This is a spontaneous process because of an entropy increase.
 - (c) This is a spontaneous process because of an entropy decrease.
 - (d) This is a spontaneous process because it is exothermic.
9. When an ideal gas is allowed to expand isothermally, which one of the following is true?
- (a) $q = 0$
 - (b) $w = 0$
 - (c) $E = 0$
 - (d) $q = -w$
10. The rate of a chemical reaction is determined by
- (a) the equilibrium constant
 - (b) the rate-determining or slow step of the mechanism
 - (c) the reaction vessel pressure
 - (d) the intermediates formed in the first step

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Sulfurous acid is a weak acid, while sulfuric acid is a much stronger acid because' is 'sulfuric acid has more oxygen atoms in its formula'.
2. True or False: The correct answer to 'Chemical reactions can be classified as either heterogeneous or homogeneous. Which of the following equations below is best classified as...' is ' $\text{C(s)} + \text{H}_2\text{O(g)} \rightarrow \text{H}_2\text{(g)} + \text{CO(g)}$ '.
3. True or False: The correct answer to ' $\text{SO}_2\text{Cl}_2 \rightarrow \text{SO}_2\text{(g)} + \text{Cl}_2\text{(g)}$ At 600 K, SO_2Cl_2 will decompose to form sulfur dioxide and chlorine gas via the above equation. If the reaction...' is 'Increasing the temperature at which the reaction occurs'.
4. True or False: The correct answer to ' $\text{S(s)} + \text{O}_2\text{(g)} \rightarrow \text{SO}_2\text{(g)}$ ' is 'Q must be close to 1.0 since there is one mole of gas on each side of the equation'.
5. True or False: The correct answer to 'What is the molarity of a sodium hydroxide solution that requires 42.6 mL of 0.108 M HCl to neutralize 40.0 mL of the base?' is '0.115 M'.
6. True or False: The correct answer to 'Each resonance form of the nitrate ion, NO_3^- , has how many sigma and how many pi bonds?' is '1 sigma and 1 pi'.
7. True or False: The correct answer to ' $\text{S(s)} + \text{O}_2\text{(g)} \rightarrow \text{SO}_2\text{(g)}$ ' is 'one mole of sulfur atoms reacts with one mole of oxygen molecules to yield one mole of sulfur dioxide molecules'.
8. True or False: The correct answer to 'Approximately how many half-lives will it take to reduce 10.0 kg of a radioactive substance to 1.0 microgram of that substance?' is '13'.
9. True or False: The correct answer to 'When the dichromate ion is reacted, one of its most common products is Cr^{3+} . What is the oxidation state (oxidation number) of chromium...' is '6+, reduction'.
10. True or False: The correct answer to 'How many electrons, neutrons, and protons are in an atom of Cr?' is '27 electrons, 27 protons, 24 neutrons'.

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